

Pat Aloo

Software Developer

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WEBSITES, PORTFOLIOS, PROFILES

Github

<https://github.com/pat-liemo>

LinkedIn

<https://www.linkedin.com/in/pat-aloo/>

PROFESSIONAL SUMMARY

Software Engineer building full-stack, mobile, and spatial-computing products — from an AI-powered automation platform integrating Meta and Shopify APIs to a LiDAR-based 3D capture pipeline with native ARKit. Comfortable across React, Flask, Kotlin, and Swift, with a bias toward shipping and testing real, working systems.

EDUCATION

Sep 2021 – Jul 2026 Nairobi, Kenya	Bachelor of Science, Computer Science <i>Kenyatta University</i>
Sep 2023 – Mar 2024 Nairobi, Kenya	Software Engineering <i>Moringa School</i>
2017 – 2021 Thika, Kenya	Kenya Certificate of Secondary Education(KCSE) <i>MaryHill Girls' High School</i> <i>A- (78 Points)</i>

PROFESSIONAL EXPERIENCE

May 2026 – Current Baba Dogo, Nairobi	Software Developer Intern <i>Shopzetu</i> • Built responsive and interactive user interfaces using React and JavaScript. • Developed backend services and RESTful APIs using Flask and Python. • Designed and integrated relational databases to support application functionality and data management. • Integrated third-party APIs and external services into web applications.
Jun 2024 – Aug 2024 Nyali, Mombasa, Kenya	Software Developer Intern <i>Brance Technologies Limited</i>  • Designed and implemented backend modules in PHP following RESTful architecture principles. • Improved database query performance through query restructuring and indexing key tables. • Participated in code reviews and agile sprint meetings, contributing to cleaner architecture and faster delivery cycles. • Collaborated directly with clients to translate requirements into technical specifications.
Jul 2023 – Sep 2023 Upper hill, Nairobi, Kenya	Information Technology Intern <i>Flag 42</i>  • Tested scalable application features aligned with evolving product requirements.

- Strengthened code quality by incorporating **unit testing** and debugging best practices.

LANGUAGES/Frameworks/Tools/Concepts

Languages:

JavaScript, Python, PHP, Kotlin, SQL, HTML/CSS (Bootstrap, Tailwind)

Frameworks/Libraries:

React, Flask, SQLAlchemy, Jetpack Compose, Three.js

Tools:

Git, Docker, Linux (Ubuntu), WebXR

Concepts:

OOP, REST APIs, GraphQL, Database Design, Data Structures and Algorithms

PROJECTS

Aug 2025 – Present

SCOUTR

360° Virtual Tour & Digital Twin Platform

- Built a full-stack proptech platform: **React/Three.js web tour viewer**, **React Native/Expo mobile field-capture app**, and **PostgreSQL backend**.
- Engineered a **LiDAR 3D pipeline** — Apple RoomPlan capture, custom USDZ→glTF conversion, and an original photogrammetric texture-projection algorithm mapping real 360° photography onto 3D mesh geometry.
- Built sensor-fusion position tracking (compass/pedometer dead-reckoning) plus a native **Swift/ARKit** module for spatial capture without GPS.

Stack: React, Three.js, React Native, Swift/ARKit, Python/FastAPI, PostgreSQL

Jun 2026 – Sep 2026

SOCIAL MEDIA AI MANAGER

AI-Powered Social Commerce Automation Platform

- Engineered a full-stack AI customer-support platform integrating **Meta Graph API, Instagram, and Shopify**.
- Implemented **real-time Meta webhooks** for social messages and comments.
- Built an **AI intent-classification pipeline** for product, inventory, order, shipping, and complaint inquiries.
- Developed **human-in-the-loop escalation** for complaints, automatically disabling AI and notifying support agents.

Stack: React, Flask, Shopify API, Anthropic API, PostgreSQL, Meta API

Sep 2025 – Jan 2026

FINANCE TRACKING APP (Android)

Personal Finance Tracking Mobile App

- Built a personal finance tracker on **Kotlin + Jetpack Compose**, using an **MVVM architecture** with ViewModel/StateFlow to drive reactive UI updates as transactions are added or edited.
- Designed a **Room (SQLite)** persistence layer with a relational schema for transactions, categories, and budgets, enabling structured expense categorization and real-time balance/spending calculations.
- Implemented **Material 3** theming with dynamic color and adaptive layouts, plus custom Compose components (charts/summary cards) for at-a-glance spending breakdowns by category.

Stack: Kotlin, Jetpack Compose, Room, Coroutines/Flow, Material 3

CERTIFICATIONS & COURSES

Harvard's CS50 Understanding Technology

2023

Moringa's Software Engineering

2024

IBM AI Fundamentals | IBM Skills Build

2025